Probabilistic School Bus Routing with Stop Selection

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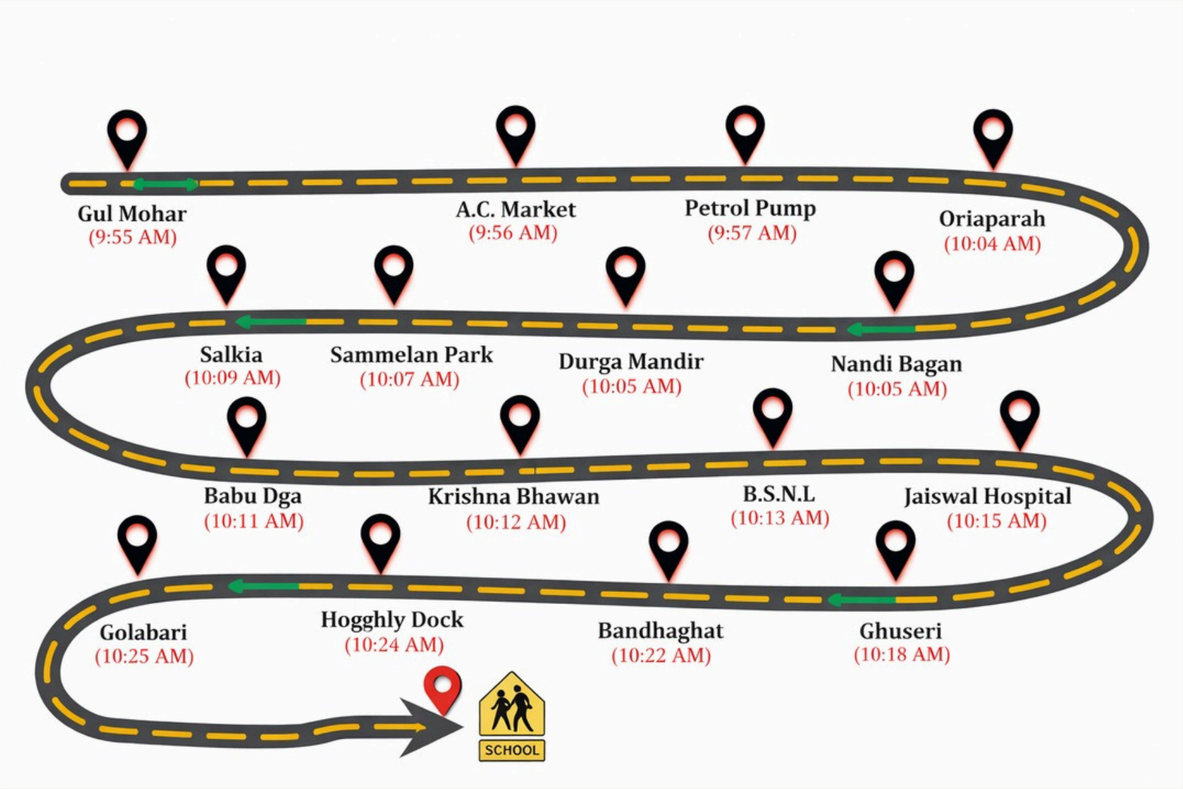

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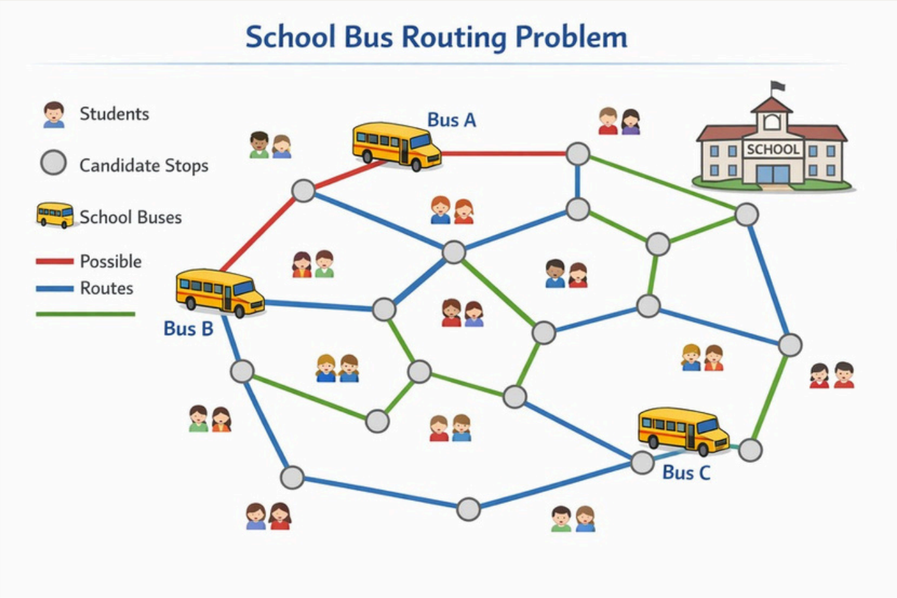
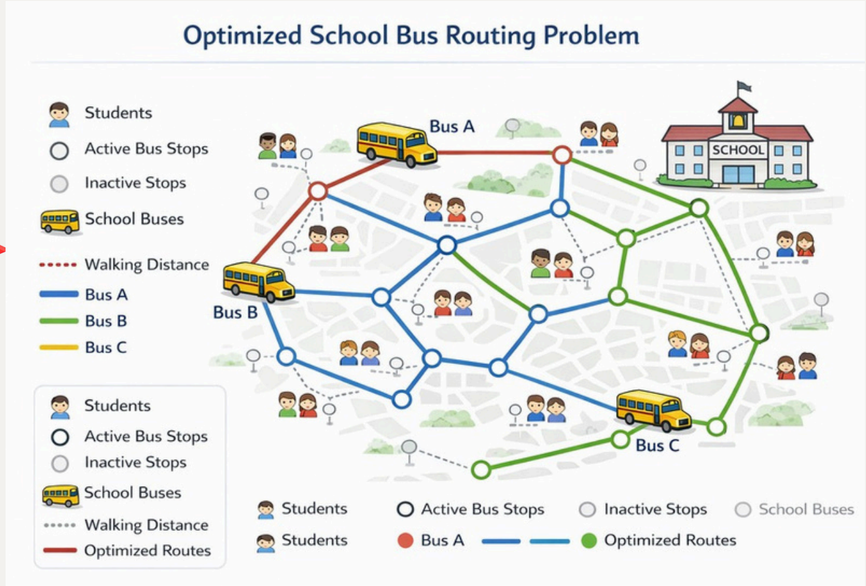
Project Context and Motivation

- **ApplicationDomain:**IntelligentTransportationSystem(ITS) [school bus - routing]

- **Real world context:** Student assignment to stops, route sequencing and traffic variability

- **Reason for uncertainty:** random daily demand, spatial distribution and traffic speed fluctuations





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Project Objective and Scope

Exact Objective:

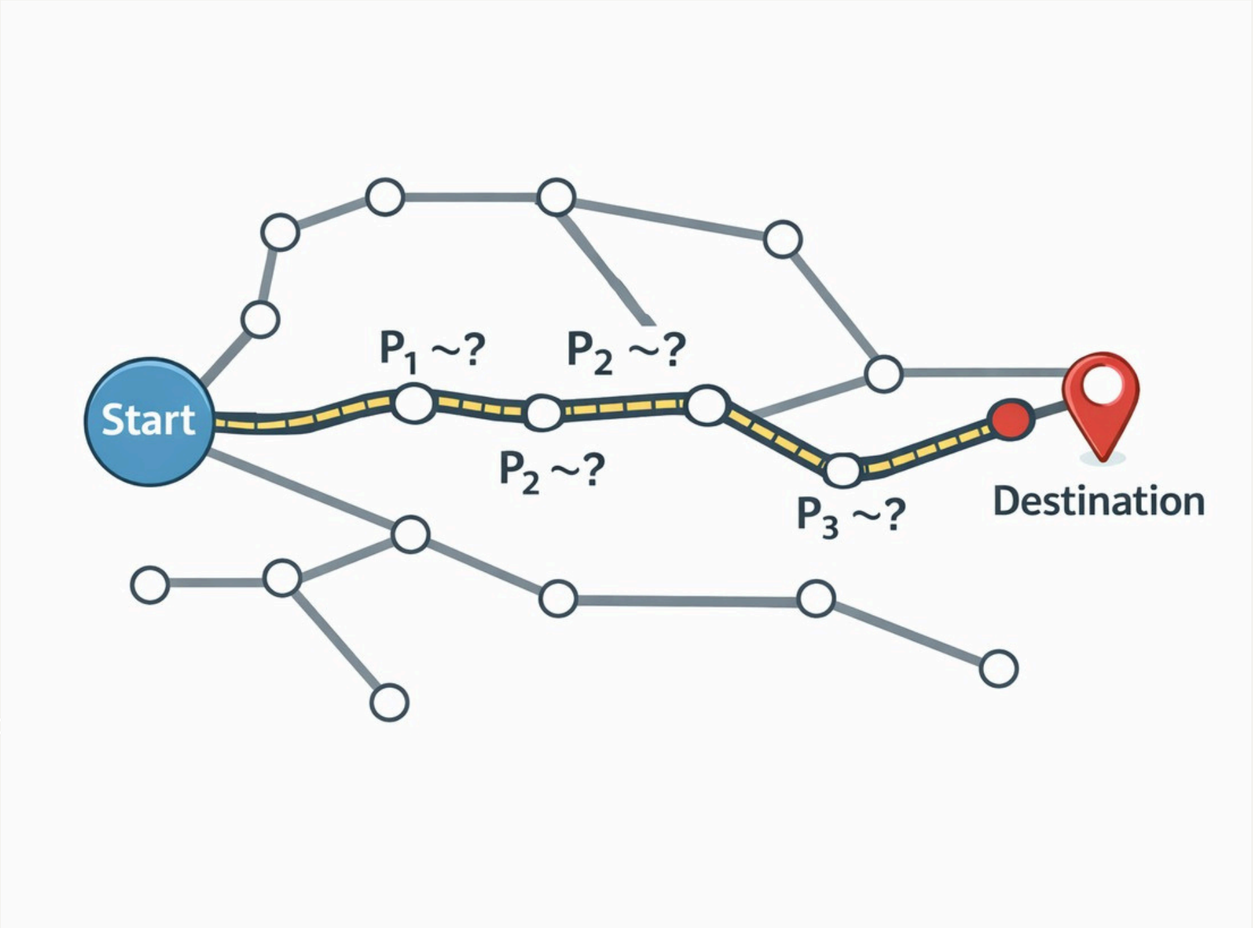
- Minimize expected travel time ofall school buses, **minE[Troute]**

Decision Task:

- Select stops, asign students and sequence routes

Scope:

- Static road network, stochastic demand and traffic

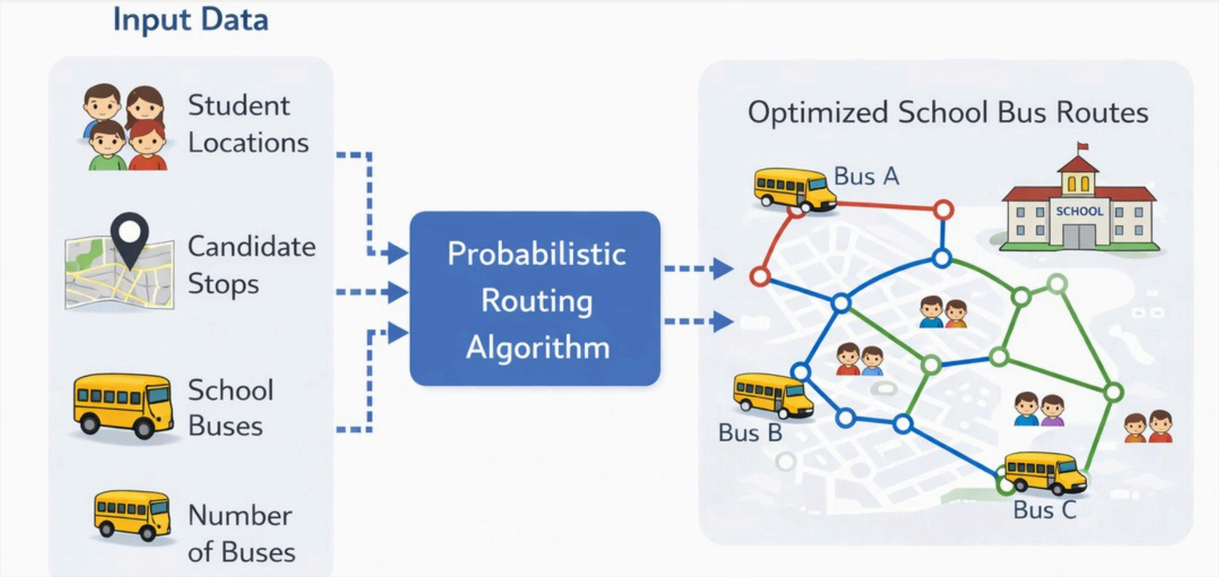


Project System Overview

- **Pipeline:**

Sample demand N~Poisson(λ) → ***Generate spatial locations*** → ***Students allocation*** → ***Route construction*** → ***EDA based route sampling*** → ***Travel time computation*** → ***Monte Carlo Simulation***

- **Components:** Student-stop assignment,EDA- based routing



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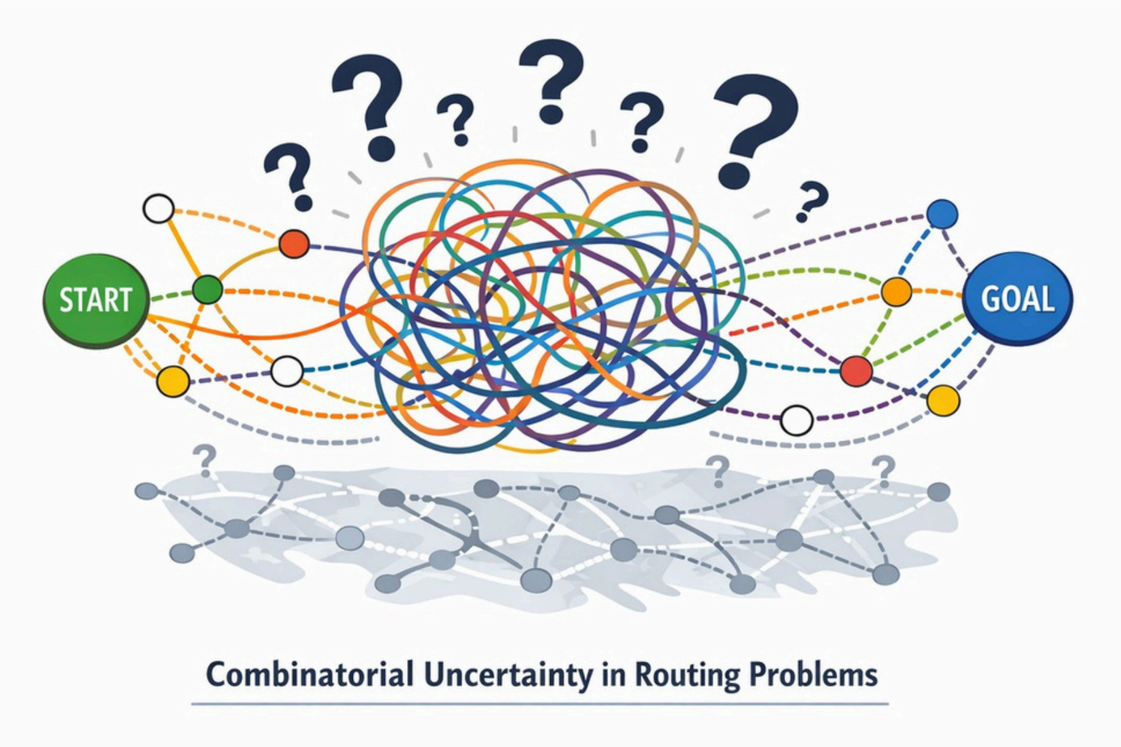
Source of Uncertainty in the Project

- Student demand variability
- Student spatial randomness
- Traffic speed variability
- Model assumption uncertainty
- Unmodeled delay components



Key Random Variables and Their Roles

1. ***Student Demand :*** $N \sim Poisson(\lambda)$
2. ***Student Spatial Location:*** $X_i = (X_i, Y_i)$
3. ***Walking distance:*** $W_{ij} = \|X_i - S_j\|$
4. ***Traffic Speed:*** $V \sim Lognormal(\mu_v, \sigma_v^2)$
5. ***Route Travel Time:*** $T_{route} = D/V$
6. ***Total System Travel Time (Multiple buses):*** $T_{total} = \sum_{k=1}^{B} T_k$

Probabilistic Models and Assumptions

- ***Probability distributions or stochastic models used used:***
  - $N \sim Poisson(\lambda)$
  - $X_i \sim Uniform(\Omega)$
  - $V \sim Lognormal(\mu_v, \sigma_v^2)$
  - $T_{route} = D/V$
  - EDA: Probabilistic model over route permutations
- ***Independence/dependence assumptions:*** *Student locations assumed independent,Demand independent of traffic speed,Walking distance depends on spatial location,Route times may include covariance terms,EDA assumes marginal factorization of permutations*
- ***Justification grounded in project behavior:*** *Reflects arrival randomness,Captures spatial uncertainty,Models skewed traffic behavior,Balances realism & tractability*

RISK BOUNDING:

Since T(route)>=0:

MARKOV INEQUALITY:

$$P(T_{route} \geq a) \leq E[T_{route}]/a$$

CHEBYSHEV INEQUALITY:

$$(|T_{route} - \mu| \geq \varepsilon) \leq Var(T_{route}) / \leq \varepsilon^2$$

these provide mathematical guarantees for extreme delays.

MONTE CARLO ESTIMATION

Expectation estimated by:

$$\hat{\mu} = \frac{1}{M} \sum_{m=1}^{M} T_{route}^{(m)}$$

By Weak Law of Large Numbers:

$$\hat{\mu} \xrightarrow{P} E[T_{route}]$$

Monte Carlo allows:

Estimating mean

Estimating variance

Studying tail risk

Model - Implementation Alignment

Implementation steps:

1. sample demand N.
2. sample student locations xi.
3. Assign via walking distance Wij.
4. Generate routes using EDA: $p(R) = \prod p(S_{j_i})$
5. sample traffic speed V.
6. Compute T(route).

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Current Strength and Limitations

Strengths:

1. Full stochastic formulation.
2. Closed-form expectation.
3. Covariance structure.
4. Monte Carlo validation.

Limitations:

1. Scalability for large N.
2. independence assumption in EDA.
3. Delay components not modeled.
4. Distribution validation pending.

Future Extensions and Final Framework

Extension:

Add delay variable: $T_{route} = \frac{D}{V} + \Delta$

Risk-based objective: $\min E[T_{route}] + \alpha Var(T_{route})$

Final Framework:

Random Demand:

Spatial Randomness

Lognormal Speed

↓

Random Travel Time

↓

Monte Carlo Estimation

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Risk-Aware Optimization

We move from deterministic routing to probabilistically grounded decision-making

THANK YOU